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Paradise Lost? Estimating the Economic Impact of Hawaii's Mandatory Quarantine using Synthetic Historical Controls

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Summary This paper uses a novel causal time series method to estimate the causal impact of Hawaii's quarantine policy to combat the spread of COVID-19 on Hawaii's tourism industry.

Keywords: *Synthetic Controls, Policy Evaluation, COVID-19*

1. INTRODUCTION

Many of the most pressing policy questions in economics concern one-time, systemic shocks: What would the United States' economy have looked like absent the revolution? What would have happened to unemployment in Great Britain without Brexit? In these cases, credible untreated units often do not exist, making standard causal inference tools, like difference-in-differences (DiD) and synthetic control (SCM), difficult or impossible to apply.

This paper takes up one such case: Hawaii's near-total quarantine in March 2020. While most regions experienced economic contraction from COVID-19 through voluntary distancing, fear, or uneven mandates, Hawaii implemented one of the few deliberate shutdowns in the U.S.: a policy aimed not merely at mitigation of the virus by agreement, but by actively suppressing travel and commerce. On March 26, 2020, the state imposed a 14-day quarantine on all arrivals, effectively sealing its borders. Hawaii's economic and tourism collapse was thus, in part, an intended consequence. Within a month, a third of the state's labor force was unemployed Ivanova (2020).

Quantifying economic impacts of such an intervention poses serious challenges. As noted by Callaway and Li (2023) and Bilgel (2022), the assumptions underlying DiD and SCM are often violated in COVIS-19 settings.[1] In a global pandemic, there are no unaffected units. Hawaii is geographically and economically distinct, and every plausible control — whether U.S. state or island nation — faced concurrent shocks. In such a setting, there is no valid cross-sectional counterfactual.

To address this, I apply the Synthetic Historical Control (SHC) method introduced by

[1]Of course, this is not universal. Researchers have productively applied SCM and related panel methods to study the effects of non-pharmaceutical interventions during the pandemic, especially when some entities implemented policies like mask mandates or lockdowns while others did not Yang (2021); Friedson et al. (2021); Gibson and Sun (2023); Breidenbach and Mitze (2021); Bulle et al. (2022); Ke and Hsiao (2022, 2023). These comparative case study designs work well when variation in treatment exists across geography.

Chen et al. (2024), which constructs a counterfactual using only the treated unit's own pre-intervention history. SHC represents a fundamental shift in the SCM paradigm: rather than borrowing strength across space, it borrows strength across time. This allows us to recover counterfactual outcomes when the treated unit is one-of-a-kind and no credible donors exist, exactly the situation in Hawaii.

This paper contributes to the literature on the broader economic consequences of lockdown-style policies—beyond their direct effects on infection or mortality. Understanding the economic costs of aggressive containment measures is vital for future health policy, particularly in tourism-dependent regions where travel restrictions can have outsized effects. Hawaii's case offers a uniquely compelling setting: its geographic isolation, centralized governance, and heavy reliance on tourism allowed it to pursue a form of near-total border control that was effectively impossible in mainland states. As such, Hawaii provides an empirical glimpse into a "zero-COVID"-like strategy implemented within the U.S. federal system—what might be called Great Isolation with Aloha characteristics.

Methodologically, this is the first real-world policy application of the SHC approach outside of its original proposal. More broadly, this paper illustrates how SHC opens a path forward for causal inference in settings where standard panel methods are unusable — including many of the policy shocks economists most want to study.

2. METHODOLOGY

We know what happened given that Hawaii mandated two-week quarantines. We do not know what would have happened had they not done this. Given its novelty and radical deviation from SCM, this section summarizes the SHC method by Chen et al. (2024). For a comprehensive treatment of the method, including the theoretical proofs and asymptotic properties, readers are referred to Chen et al. (2024).

Notation Let $\mathbb{R}$ denote the set of real numbers, and let $\|\cdot\|_1$ denote the usual ℓ_1 vector norm. Throughout, I denote sets using calligraphic letters (e.g., $\mathcal{T}, \mathcal{S}, \mathcal{N}$) for clarity. Scalars are represented using plain lowercase letters (e.g., h, t, n, m), and matrices are denoted by bold uppercase letters (e.g., $\mathbf{X}, \mathbf{W}, \mathbf{Y}$). Given a vector $\mathbf{x} = (x_1, x_2, \ldots, x_T)^\top \in \mathbb{R}^T$ and an index set $\mathcal{I} \subseteq \{1, \ldots, T\}$, I write $\mathbf{x}_\mathcal{I} := (x_t)_{t \in \mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$ to denote the subvector of $\mathbf{x}$ corresponding to indices in $\mathcal{I}$, preserving their original order.

Let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length, i.e., the number of periods used to construct the SHC match. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. To reconstruct the evaluation window, we compare it against earlier segments of the treated unit's own pre-treatment trajectory.

Define the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$ as a collection of N overlapping length-m subvectors extracted from the smoothed pre-treatment series. Each column is a contiguous subvector of the form $\widetilde{\mathbf{y}}_{[i,i+m-1]}$, with i ranging over eligible start points in $\mathcal{T}_1$ that precede the evaluation window. The precise construction is given in Section 2.2.

2.1. Synthetic Historical Control Method

Potential Outcomes Following the Rubin potential outcomes framework, let y_t^1 and y_t^0 denote the potential outcomes for the treated unit at time $t \in \mathcal{T}$, corresponding to the outcomes under treatment and no treatment, respectively. The observed outcome at time t is $y_t = d_t y_t^1 + (1 - d_t) y_t^0$, where $d_t = \mathbf{1}\{t > T_0\}$.

The fundamental problem of causal inference is that for $t > T_0$, we observe only the treated potential outcome $y_t^1 = y_t$, while the untreated potential outcome y_t^0 remains unobserved. Let the SHC estimator for y_t^0 be denoted $\widehat{y}_t^0 := y_t^{\text{SHC}}$ for all $t \in \mathcal{T}_2$. Define the observed post-treatment outcome vector as $\mathbf{y}_{\text{post}} := (y_t^1)_{t \in \mathcal{T}_2}$ and the estimated counterfactual vector as $\mathbf{y}_{\text{post}}^0 := (\widehat{y}_t^0)_{t \in \mathcal{T}_2}$. The out-of-sample treatment effect is

$$\alpha_t := y_t^1 - y_t^0, \quad \text{for } t \in \mathcal{T}_2, \tag{2.1}$$

with estimator $\widehat{\alpha}_t := y_t - \widehat{y}_t^0$. Stacking over $t \in \mathcal{T}_2$ yields:

$$\widehat{\boldsymbol{\alpha}} := \mathbf{y}_{\text{post}} - \mathbf{y}_{\text{post}}^0. \tag{2.2}$$

The average treatment effect on the treated (ATT) is then given by:

$$\widehat{\text{ATT}} := \frac{1}{n} \sum_{t=T_0+1}^{T_0+n} \left(y_t - \widehat{y}_t^0 \right) = \frac{1}{n} \mathbf{1}^\top \widehat{\boldsymbol{\alpha}}, \tag{2.3}$$

where $\mathbf{1} \in \mathbb{R}^n$ is a vector of ones.

2.2. The Model

The SHC method introduced by Chen et al. (2024) constructs a counterfactual using the treated unit's own pre-treatment time series.[2] SHC requires a few assumptions: Assumption 2.1, Assumption 2.2, and Assumption 2.3.

Assumption 2.1. (Error Properties) *The error term satisfies* $\mathbb{E}[\varepsilon_t] = 0$ *and* $\mathbb{E}[\varepsilon_t^2] < \infty$ *for all* $t \in \mathcal{T}$.

[2]SCM estimates the counterfactual by solving:

$$\mathbf{w}^* = \underset{\mathbf{w} \in \Delta^{N_0}}{\text{argmin}} \, \|\mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w}\|_2^2, \quad \forall t \in \mathcal{T}_1, \tag{2.4}$$

where $\mathbf{Y}_0 \in \mathbb{R}^{T_0 \times N_0}$ is the donor matrix of unexposed units and $\Delta^{N_0} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} : \|\mathbf{w}\|_1 = 1 \right\}$ is the probability simplex.

Assumption 2.1 says unmodeled fluctuations or noise in the outcome are random.

Assumption 2.2. (Latent Component Smoothness and Matching Condition) *The latent trend $\widetilde{y}(t)$ is $(H+1)$ times differentiable for some integer $H \geq 3$, with derivative bounded as $\left|\widetilde{y}^{(H+1)}(t)\right| < \varepsilon$ uniformly over $t \in \mathcal{T}_1$. Additionally, $\exists \mathbf{w}^* \in \Delta^{N-1}$ such that $\mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*$.*

Assumption 2.2 allows us to approximate the post-treatment latent component using a local linear extrapolation of pre-treatment values and posits that perfect match exists within the donor pool for the evaluation window, much as Abadie et al. (2010) assume about donor pool units matching to the treated unit.

Assumption 2.3. (Feasibility) *There exists a weight vector $\mathbf{w} \in \Delta^{N-1}$ such that $\widehat{\mathbf{y}}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}$, with $\|\mathbf{w}\|_0 \leq s \ll N$. Let $\widetilde{\mathbf{Y}}^{(s)}_{pre}$ denote the submatrix formed by the active columns of $\widetilde{\mathbf{Y}}_{pre}$ indexed by* $\operatorname{supp}(\mathbf{w})$*; then its Gram matrix satisfies $\left(\widetilde{\mathbf{Y}}^{(s)}_{pre}\right)^\top \widetilde{\mathbf{Y}}^{(s)}_{pre} \succ 0$.*

Chen et al. (2024) note that Assumption 2.3 is the sample-variant of Assumption 2.2. It says that the SHC weights are identifiable, sparse, and numerically stable, similar to the assumptions described by Abadie (2021).

The first step of the SHC method smooths $\mathbf{y}_{\text{pre}}$ using local linear regression, where we fit the regression model:

$$y_j \approx a_t + b_t(j-t), \quad \text{for } j = 1, \ldots, T_0, \tag{2.5}$$

with local intercept $a_t \in \mathbb{R}$ and slope $b_t \in \mathbb{R}$. Define the centered time vector $\mathbf{s}_t = (1-t, 2-t, \ldots, T_0-t)^\top \in \mathbb{R}^{T_0}$, and the design matrix $\mathbf{X}_t = [\mathbf{1}\ \mathbf{s}_t] \in \mathbb{R}^{T_0 \times 2}$. The diagonal kernel weight matrix is:

$$\mathbf{W}_t = \operatorname{diag}(w_{1t}, w_{2t}, \ldots, w_{T_0 t}), \quad w_{jt} \propto \exp\left(-\frac{1}{2}\left(\frac{j-t}{h}\right)^2\right), \tag{2.6}$$

where $(h > 0$ is a bandwidth parameter governing smoothness. The local coefficients $\boldsymbol{\beta}_t = (a_t, b_t)^\top$ solve quadratic programming problem that is strictly convex

$$\boldsymbol{\beta}_t = \underset{\boldsymbol{\beta} \in \mathbb{R}^2}{\operatorname{argmin}} \left\| \mathbf{W}_t^{1/2} \left(\mathbf{y}_{\text{pre}} - \mathbf{X}_t \boldsymbol{\beta}\right) \right\|_2^2. \tag{2.7}$$

Fortunately, this optimization admits a closed-form solution for the coefficients:

$$\boldsymbol{\beta}_t = \left(\mathbf{X}_t^\top \mathbf{W}_t \mathbf{X}_t\right)^{-1} \mathbf{X}_t^\top \mathbf{W}_t \mathbf{y}_{\text{pre}}. \tag{2.8}$$

The smoothed outcome at time t is $\widetilde{y}_t = a_t = \mathbf{e}_1^\top \boldsymbol{\beta}_t$, where $\mathbf{e}_1 = (1,0)^\top$. Let $\widetilde{\mathbf{y}} = (\widetilde{y}_1, \ldots, \widetilde{y}_{T_0})^\top \in \mathbb{R}^{T_0}$ denote the full smoothed series. After we have smoothed over the pre-treatment period before the evaluation period, we build our donor matrix of historical control segments/units.

In the second step, SHC constructs $N = T_0 - m - n + 1$ overlapping temporal segments of length m. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} := \left[\widetilde{\mathbf{y}}_{[1,m]} \quad \widetilde{\mathbf{y}}_{[2,m+1]} \quad \cdots \quad \widetilde{\mathbf{y}}_{[N,N+m-1]}\right] \in \mathbb{R}^{m \times N}$.

Example 2.1. (Constructing $\widetilde{\mathbf{Y}}_{\text{PRE}}$) *Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months in order to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is: $N = T_0 - m - n + 1 = 60 - 12 - 6 + 1 = 43$. Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.*

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} \quad & \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1 \end{aligned} \tag{2.9}$$

Chen et al. (2024) note that in the general case when we have more donor units than we do the length of the segments ($N > m$), the optimization becomes an ill-posed problem. The regularization term stabilizes the optimization by penalizing weight allocations along directions of low variance in the donor pool, reducing sensitivity to collinear segments. Formally, $\mathbf{C}_0$ contains the eigenvectors of the Grammian matrix $\mathbf{G} \in \mathbb{R}^{N \times N}$. $\mathbf{C}_0$ corresponds to the eigenvalues less than $\tau > 0$. These directions capture low-variance combinations of donor segments, and the penalty shrinks weights in those directions.[3]. The corresponding weight vector $\mathbf{w}^*$ is used to compute the counterfactual by applying it to the shifted donor segments:

$$\widetilde{\mathbf{Y}}_{\text{post}} = \left[\widetilde{\mathbf{y}}_{[1+m,m+n]} \quad \widetilde{\mathbf{y}}_{[2+m,m+n+1]} \quad \cdots \quad \widetilde{\mathbf{y}}_{[N+m,N+m+n-1]}\right] \in \mathbb{R}^{n \times N}. \tag{2.10}$$

This yields in-sample and out-of-sample predictions for the SHC estimator. The full counterfactual vector is then: $\mathbf{y}^0 = \widetilde{\mathbf{Y}} \mathbf{w}^* \in \mathbb{R}^{m+n}$.

Bandwidth Selection via Cross-Validation To select the bandwidth parameter h used in local linear smoothing of $\mathbf{y}_{\text{pre}}$, I implement a *leave-one-out cross-validation* (LOOCV) procedure over a grid of candidate values $h \in [h_{\min}, h_{\max}] \subset \mathbb{R}_{>0}$. This procedure chooses the bandwidth that minimizes out-of-sample prediction error for the smoothed pre-treatment series $\widetilde{\mathbf{y}} \in \mathbb{R}^{T_0}$, as defined in Equation~(2.8).

For each candidate h, and for each time point $t \in \mathcal{T}_1$, define Gaussian kernel weights

$$w_{\tau t}^{(h)} = \exp\left(-\frac{1}{2}\left(\frac{\tau - t}{h}\right)^2\right), \quad \tau \in \mathcal{T}_1 \setminus \{t\}, \tag{2.11}$$

used to construct the diagonal matrix $\mathbf{W}_t^{(-t)} \in \mathbb{R}^{(T_0-1)\times(T_0-1)}$, omitting the index t. Let $\mathbf{X}_t^{(-t)} \in \mathbb{R}^{(T_0-1)\times 2}$ and $\mathbf{y}_{\text{pre}}^{(-t)} \in \mathbb{R}^{T_0-1}$ denote the design matrix and response vector with time t excluded.

Then the local linear coefficients $\boldsymbol{\beta}_t^{(-t)} = (a_t^{(-t)}, b_t^{(-t)})^\top$ are obtained by solving Equation~(2.7). The leave-one-out smoothed value at time t is then:

$$\widetilde{y}_t^{(-t)} = a_t^{(-t)} = \mathbf{e}_1^\top \boldsymbol{\beta}_t^{(-t)}, \quad \text{where } \mathbf{e}_1 = (1, 0)^\top. \tag{2.12}$$

[3]The ς (sigma) term is a small positive constant used to stabilize the solution in cases of near multicollinearity.

To pick the ideal bandwidth, we minimize:

$$h^* = \underset{h \in [h_{\min}, h_{\max}]}{\operatorname{argmin}} \left\{ \mathrm{CV}(h) := \frac{1}{T_0} \sum_{t=1}^{T_0} \left(y_t - \widetilde{y}_t^{(-t)} \right)^2 \right\}. \tag{2.13}$$

Once selected, the optimal bandwidth h^* is used to construct the full smoothed pre-treatment series.

Control Group Selection Chen et al. (2024) note that we should have a parsimonious approach to which historical controls we use, rather than choosing all of them. In their examples, SHC typically requires less than 28 controls. They advocate for a forward selection procedure to choose the optimal control group. I implement a slightly modified variant of this method to choose the ideal number of historical controls. Let $\mathcal{N} = \{1, 2, \ldots, N\}$ index the full set of donor segments in the pre-treatment matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$. At each step the algorithm selects the donor index $i_j \in \mathcal{N} \setminus \mathcal{S}_{j-1}$ that, when added to the current donor set $\mathcal{S}_{j-1}$, minimizes the in-sample mean squared error (MSE) between the evaluation window and in-sample fit:

$$\mathcal{S}_j = \mathcal{S}_{j-1} \cup \left\{ \underset{i \in \mathcal{N} \setminus \mathcal{S}_{j-1}}{\operatorname{argmin}} \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S}_{j-1} \cup \{i\})} \mathbf{w}^{(j)} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\}, \quad \mathcal{S}_0 = \emptyset, \tag{2.14}$$

where $\widetilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S})}$ denotes the submatrix of $\widetilde{\mathbf{Y}}_{\text{pre}}$ with columns indexed by $\mathcal{S}$, and the weights $\mathbf{w}^{(j)} \in \Delta^j$ are the optimal simplex-constrained weights for the selected donor set $\mathcal{S}_j$. This procedure greedily expands the donor set one segment at a time, always choosing the segment that yields the greatest marginal reduction in in-sample error.

To avoid overfitting, Chen et al. (2024) manually choose the number of donors the forward selection procedure should choose. In their case, they experiment with the single best historical segment (which rarely does well), the best 15, 20, and 27 historical control units. However, this is a subjective decision that will yield different results depending on what we select. To mitigate researcher degrees of freedom, I view the number of historical segments as a tuning parameter. I employ a forward selection-based algorithm to choose the number of controls. For my implementation of SHC, I employ an early stopping rule based on a modified Bayesian information criterion (BIC), borrowing inspiration from Shi and Huang (2023). For each donor set size j, I compute:

$$\mathrm{BIC}(j) = m \cdot \log\left(\mathrm{MSE}_j\right) + \lambda j, \tag{2.15}$$

where MSE_j is the in-sample error using donor set $\mathcal{S}_j$, and $\lambda = \log(m)$ penalizes the inclusion of too many donors. The selection procedure terminates when $\mathrm{BIC}(j)$ increases for two consecutive steps. The final donor set $\mathcal{S}^*$ is then taken to be $\mathcal{S}_{j^*}$, where:

$$j^* = \underset{j}{\operatorname{argmin}}\ \mathrm{BIC}(j). \tag{2.16}$$

This approach adaptively selects a sparse donor subset that closely matches the treated unit's pre-intervention trajectory. The selected donor set $\mathcal{S}^*$ is used to compute the SHC model predictions. I would like to mention that this forward selection method holds much in common with recently proposed methods from the literature. For example, Shi and Huang (2023) propose the forward selected panel data approach which selects the comparison group for the panel data approach by Hsiao et al. (2012) using an aggressive forward selection procedure. Li (2024), correcting for overfitting in the former, proposed the forward Difference-in-Differences estimator, where a forward selection approach is used to select the donor pool for the DiD model. Similar advances have been made SCM, as developed by Cerulli (2024), where a similar selection mechanism is used to estimate the treatment effect. It would be interesting, but is far outside the scope of this work, to compare these estimators to one another and see when their in-sample fits and point estimates agree with one another.

Inference Quantifying uncertainty around the counterfactual predictions is essential for valid inference. Chen et al. (2024) do not provide any inferential method to use aside from in-time placebos. Following recent work in post-modeling inference by Cattaneo et al. (2025), I construct agnostic conformal prediction intervals around the counterfactual series. These intervals are distribution-free, require no parametric assumptions, and are valid under exchangeability of residuals between the pre- and post-treatment periods (or, good in-sample fit).

Let $\mathbf{y}^0_{\text{eval}} \in \mathbb{R}^m$ denote the SHC-fitted values during the evaluation period and define the residuals as:

$$\boldsymbol{\varepsilon}_{\text{eval}} := \mathbf{y}_{\text{eval}} - \mathbf{y}^0_{\text{eval}}. \tag{2.17}$$

I then compute the sample mean $\bar{\varepsilon}$ and variance $\widehat{\sigma}^2$ of these residuals:

$$\bar{\varepsilon} = m^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} \varepsilon_t, \quad \widehat{\sigma}^2 = (m-1)^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} (\varepsilon_t - \bar{\varepsilon})^2. \tag{2.18}$$

To construct prediction intervals for the post-treatment counterfactual $\mathbf{y}^0_{\text{post}} \in \mathbb{R}^n$, I assume the residuals are sub-Gaussian and apply a conformal bound based on Hoeffding-type concentration inequalities. For a miscoverage rate $\alpha \in (0,1)$, the interval half-width is given by:

$$c_\alpha = \sqrt{2\widehat{\sigma}^2 \log\left(\frac{2}{\alpha}\right)}. \tag{2.19}$$

Then, the agnostic $(1-\alpha)$ prediction interval for each $t \in \mathcal{T}_2$ is given by:

$$\left[\widehat{y}^0_t + \bar{\varepsilon} - c_\alpha,\ \widehat{y}^0_t + \bar{\varepsilon} + c_\alpha\right]. \tag{2.20}$$

Stacking these yields lower and upper bound vectors:

$$\mathbf{L} := \mathbf{y}^0_{\text{post}} + \bar{\varepsilon} \cdot \mathbf{1} - c_\alpha \cdot \mathbf{1}, \quad \mathbf{U} := \mathbf{y}^0_{\text{post}} + \bar{\varepsilon} \cdot \mathbf{1} + c_\alpha \cdot \mathbf{1}, \tag{2.21}$$

which define the conformal confidence band around the counterfactual trajectory.

This approach allows finite-sample valid coverage under the assumption that the pre- and post-treatment residuals are exchangeable. Intuitively, if the SHC model fits the pre-treatment series well (i.e., residuals are small and centered), then its extrapolations into the post-treatment period are expected to carry similarly bounded uncertainty. In this sense, the residual distribution learned from the evaluation window is used to calibrate uncertainty for the out-of-sample outcomes.

These prediction intervals reflect uncertainty in the estimated counterfactual. Hence, if the observed outcome falls far outside the prediction interval, this is interpreted as evidence of a treatment effect. Unlike traditional confidence intervals around treatment effect estimates (e.g., using standard errors), the conformal intervals here do not assume a data-generating model. I use a default coverage level of 90% (i.e., $\alpha = 0.1$) in all applications.

3. DATA

To evaluate the impact of Hawaii's COVID-19 border closure on its tourism economy, I use monthly time series spanning various economic margins, focusing on four primary outcomes: two from the labor market and two from the tourism sector. All outcomes are transformed into annualized year-over-year (YoY) growth rates. This transformation not only mitigates time trends—essential for the validity of the SHC estimator Chen et al. (2024)—but also allows treatment effects to be interpreted directly as percentage point changes.

The labor market outcomes, sourced from the Federal Reserve Economic Data (FRED), cover January 1991 to December 2020. The first is employment in Hawaii's Leisure and Hospitality sector (`HILEIHN`), a proxy for tourism activity given its sensitivity to visitor demand. The second is initial unemployment insurance claims (`HIICLAIMS`), capturing monthly inflows into unemployment across the broader economy. Together, these outcomes provide complementary views of the economic fallout: `HILEIHN` reflects the stock of employed workers in a tourism-dependent sector, while `HIICLAIMS` captures more immediate disruptions to labor market flows.

The tourism-sector outcomes come from Hawaii's Department of Business, Economic Development & Tourism (DBEDT), accessed via their public API, and span January 1990 to December 2020. The first is *visitor arrival days*, the total number of person-days spent by visitors each month. This variable integrates both the volume of arrivals and length of stay, making it a direct proxy for aggregate tourism demand. The second is the *hotel occupancy rate*, defined as the share of hotel rooms sold per night—a clean, interpretable measure of capacity utilization that reflects realized demand.

Hawaii's quarantine policy took effect in late March 2020. I define the intervention to begin in March 2020, corresponding to time $t = 362$, and formalize the treatment indicator as:

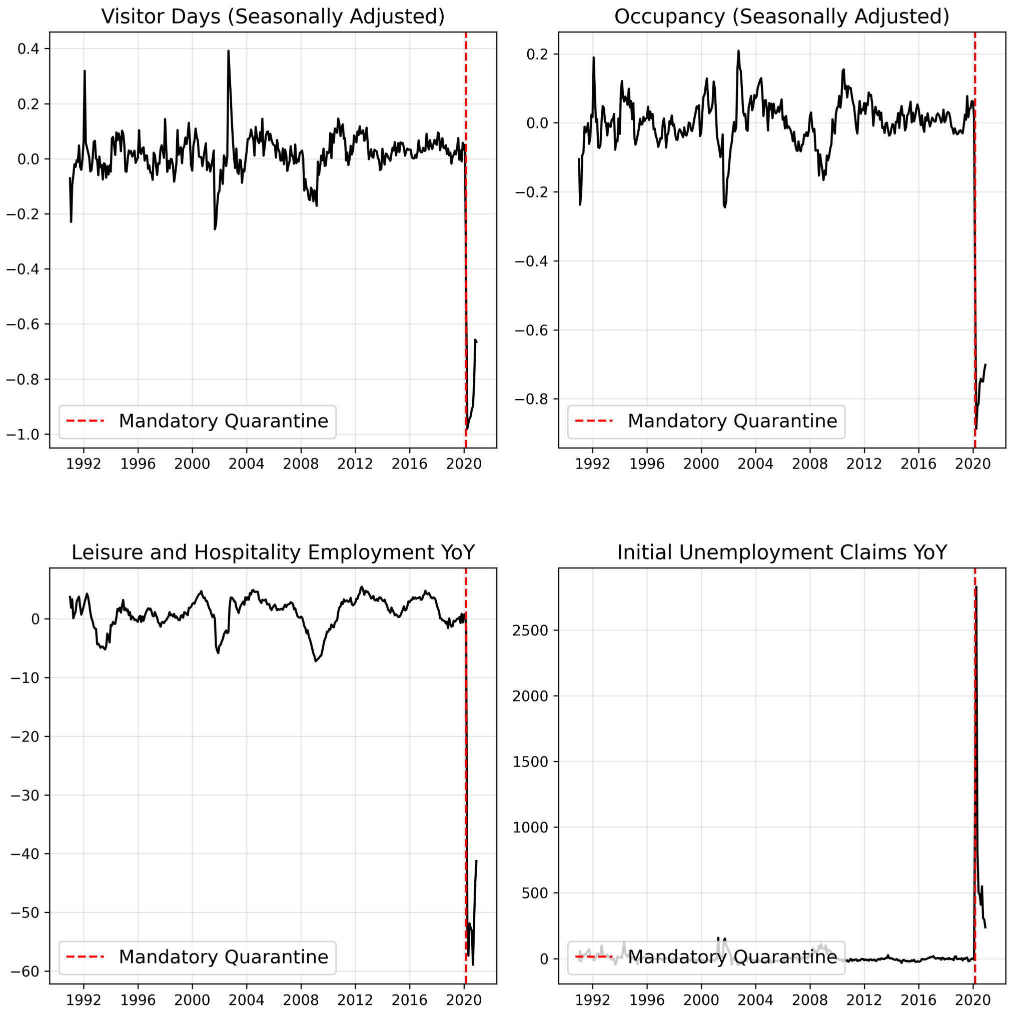
$$d_t := \mathbf{1}\{t \geq 362\},$$

so that $d_t = 1$ for all months after the policy's implementation and $d_t = 0$ otherwise. The final pre-treatment period is February 2020, yielding a pre-treatment sample of $T_0 = 361$ months.

4. RESULTS

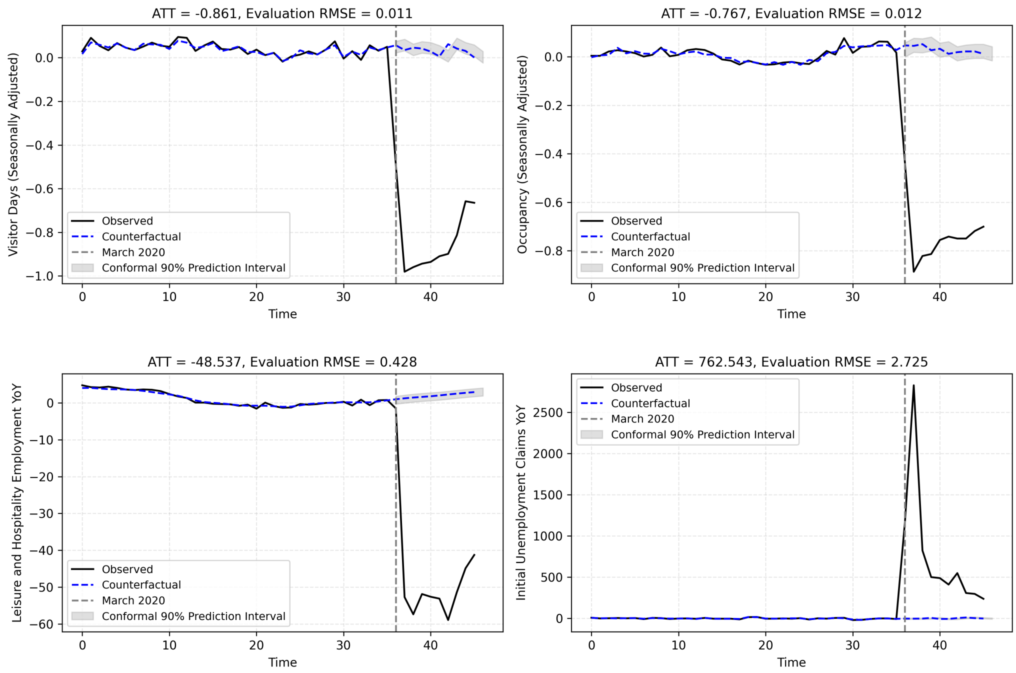
In Figure we visualize the outcomes of interest over time from January of 1991 to December 2020. We can see that the growth rate trends for all outcomes were relatively smooth across the full pre-intervention series, exhibiting seasonality and relatively low variance.

We also see that the outcomes of interest precipitously dropped when the mandatory quarantine policy was implemented (aside from unemployment claims). The plot also suggests that Assumption 2.1 and Assumption 2.2 may be valid.



Here are the initial SHC results. For these results, $m = 36$ (3 years of evaluation data), but the results remain robust up until 8 years of evaluation data.

```
/opt/hostedtoolcache/Python/3.13.5/x64/lib/python3.13/site-packages/cvxpy/problems/problem.py:
  warnings.warn(
```



The plot lists the ATT and RMSE for the outcome of interest.

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